

2017-Ph-H1-Q12

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs

From a displacement–time graph, calculate **instantaneous velocity**.

Instantaneous velocity = **gradient of the graph at a point**.

If it's a curve, draw a **tangent at that point** and calculate:

$$v = \frac{\Delta s}{\Delta t}$$

Use values from the graph to find the gradient.



Final Answer: depends on graph, but key is: **gradient = velocity**



Revision Tips:

- Gradient of:
 - **s–t graph** = velocity
 - **v–t graph** = acceleration
- Tangent = velocity at a point on a curve

